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CASE-A-THON 2026 · CASE PROMPT

Unintentional Data Poisoning

How stale and low-quality data distort decisions over time

Case sponsored by Air Combat Command, HQ ACC/ST
Questions about the case go to ajyowell@wm.edu

The student case

The problem

A logistics company uses AI to predict delivery times, estimate vehicle energy needs, prioritize work, or recommend routes. The system keeps giving answers that look reasonable. The problem is that the data behind those answers is getting worse. Old patterns no longer match how things run today. Some records stop updating. A fast, low-quality feed adds more data than a slower, more accurate one.

Your team needs to figure out how and why that data changes the algorithm's behavior over time. Then design and test a way to fix it. This is not just about cleaning a spreadsheet or spotting a bad record. The goal is to tie a data problem to a measurable change in predictions or decisions, and show whether your fix actually works.

Core question. When does more information make an AI-enabled decision process worse, and what evidence shows that your intervention restores useful performance?

Application settings

Pick one setting. Land-based delivery robots or courier logistics, aerial (drone) package delivery, or maritime logistics relevant to autonomous vessels. Possible tasks include delivery-time prediction, energy estimation, demand forecasting, or route/sequence recommendation. These are examples, not requirements. You do not need to build a vehicle, a full delivery platform, or a land-air-sea simulation.

What "unintentional data poisoning" means here

Here, the term means accidental, non-malicious decline in the data used to train, update, retrieve context for, or run an algorithm. Say which stage is affected. A frozen model fed stale inputs behaves differently from a model retrained on worse and worse records. Teams may add controlled defects in offline experiments to stand in for accidental real-world failures.

Technical note. Adversarial-machine-learning literature distinguishes intentional poisoning from evasion and other attacks. This case retains the sponsor's title but does not require an attacker or an attack demonstration. [9]

Discussion examples with related but different mechanisms

Historical representation. Reuters reported that Amazon's experimental recruiting system learned preferences from historical resumes dominated by men. Use this to discuss how data can encode bias, not as a claim that the tool independently made final hiring decisions or that its failure was caused by data age. [11]

Perception sensitivity. Researchers have shown that altered road signs can mislead vision models. The cited Tesla/Mobileye demonstration involved a speed-limit sign, while stop-sign research tested other classifiers. These deliberate input manipulations illustrate sensitivity to information quality. They are not demonstrations of accidental poisoning over time. [12, 13]

What teams must demonstrate

Requirement	Evidence expected
1. Define the decision	Identify the user, prediction or decision, consequence of error, evaluation unit, and what "better" means. Name the AI component and where information enters it.
2. Establish a reference	Build a repeatable baseline using documented source data. Explain why the evaluation outcome is credible and what uncertainty it contains. The untouched source is a reference, not automatically perfect ground truth.
3. Introduce or isolate degradation	Test stale information and low-quality information separately, then together. Record affected fields, timing, severity, source mix, and the plausible operational failure represented.
4. Show the effect over time	Use a genuine timeline or an explicitly labeled simulated progression. Show performance at multiple points, not just one before/after aggregate. Track data age, quality, and contribution volume alongside algorithm behavior.
5. Explain the mechanism	Trace the change from data to model/output to decision. Use controlled comparisons or diagnostic evidence to distinguish a plausible explanation from a story fitted to the results.
6. Demonstrate a countermeasure	Choose and implement your approach. Compare it with the unmitigated system on the same evaluation cases, including ordinary conditions and a previously unseen period or condition.
7. Identify limits and next steps	Show at least one failure, adverse tradeoff, or boundary. Explain what would justify stopping, iterating, a controlled pilot, or a larger follow-on effort. State what was learned and a bounded next action with a responsible role and decision checkpoint.

Minimum comparison structure

Test four conditions. Reference/nominal, stale only, low quality only, and combined. Run each one with and without your countermeasure. Keep the evaluation outcomes and relevant settings the same within each comparison. Teams choose the algorithm, the type of degradation, the time scale, and the fix. No method is favored.

Keep the experiment credible

"Stale" means no longer appropriate at the time of a decision, not simply downloaded from an old publication. Preserve chronological order when it exists. Do not randomly shuffle away the very change being studied. Prevent later outcomes, future records, or simulated defect labels from becoming information that the algorithm could not have known.

Keep related observations together where appropriate, such as readings from one flight or stops on one route. Disclose whether and when the model retrains. For randomized experiments, preserve settings and seeds and repeat enough runs to show whether the result is stable. A controlled simulated progression is acceptable when the dataset lacks a long operational history, but label it clearly.

Do not engineer a guaranteed win. Test whether your method keeps valid rare events, avoids throwing out useful data, and still works when the degradation changes. A well-supported negative result can score well. One experiment cannot prove universal robustness.

Deliverables and human responsibility

Submission	Expectation
Working artifact	Modest web application that demonstrates the algorithm, degradation, and countermeasure. Include setup instructions, dependencies, versions, settings, and a recorded fallback demonstration.
Results and data manifest	Machine-readable results plus dataset source/version, subset rules, dates or simulated time, split rules, modification log, and AI/prompt disclosure. Do not redistribute data without permission.
Presentation	10-minute presentations centered around the live demonstration. Your presentation should demonstrate the technical architecture while also explaining your decision-making in clear language.

Human oversight must be visible

Identify who uses the output, who reviews questionable results, when the system must defer, and how a person can correct an error. Demonstrate a brief walkthrough of one unreliable-output scenario. Show the evidence, age/quality information, explanation, and correction path the user sees.

Report human effort separately from model response time. Review, verification, correction, and escalation all count. Small peer walkthroughs may reveal usability issues, but are not proof of broad operator trust or readiness. Any participant study must follow W&M's approved process. This package asks for demonstrated oversight, not a self-awarded formal Human Readiness Level. [14]

Responsible AI and data governance

Use public, appropriately licensed, or clearly documented synthetic data. No classified, controlled, personal, or proprietary records should be uploaded for this exercise without explicit authorization. Do not attempt to reidentify people, operators, or anonymized locations. Keep credentials out of submissions. Perform experiments offline, not on vehicles or live operational services.

State who would own data quality and model performance in a pilot, how updates and failures would be logged, what approvals would be needed, and when the system should not be used. NIST's voluntary AI Risk Management Framework is an optional organizing reference, not evidence of deployment authorization. [10]

AI use and a fair workload

A small model, an existing model, or a prompt-based workflow can all qualify. You do not need to train a large model. Explain what the AI does in your workflow and evaluate it. Using AI only to write code does not count as the capability being tested.

Teams do not need to run a model live during the presentation, but organizers will ask in a short post-event survey which AI tools and models the team used.

Dataset choices

The datasets are options, not four assignments. Pick one main dataset and one manageable task, or make the case for a combination that fits. Use enough independent cases, operating variation, and time progression to back up your findings. You do not need to process an entire archive, and dataset size alone earns no points.

Option	Useful starting point	Scale and access
LaDe | land	Delivery-time estimates, demand, or service-area effects.	About 10.7 million packages, 21,000 couriers, and six months. City-level CSV files. [1, 2]
CMU drone | air	Energy-use or flight-performance estimation.	196 flights plus 13 ancillary/hover recordings. Approximately 243 MB of sensor data. [3]
MarineCadastre AIS | sea	Arrival/movement prediction or a justified logistics proxy.	Multi-year vessel-traffic archives. Select a geographic area and time span through bulk files. [6–8]
Amazon routing | land	Route sequencing or recommendation behavior learned from historical routes.	9,184 routes across five U.S. metropolitan areas. Public S3 data and schema documentation. [4, 5]

Minimum data manifest

Record the source URL, retrieval date and version, selected files, row count, independent-case count, date range, relevant fields and units, subset/filter rules, missing-value treatment, and rights statement. Preserve an untouched reference copy. Log introduced changes separately from problems already observed in the source.

Access and licenses

This package provides source pathways and suggested scopes, not copies of the full databases. Source documentation and listings were originally reviewed on 8 October 2026 and rechecked on 15 October 2026. Teams must check each dataset's license terms before use or redistribution. Organizers will post any access updates if download paths, schemas, or classroom-scale subsets change.

Dataset cards: land and air

1. LaDe: last-mile delivery operations

Dataset and files | Author paper

What is available. LaDe covers approximately 10.7 million packages handled by 21,000 couriers over six months. Delivery and pickup CSVs include event times, locations, courier identifiers, and area information. The repository lists about 2.88 GB. [1, 2]

Suggested start. Use one city's delivery CSV and a chronological subset spanning several operating periods. Predict delivery duration from task acceptance, or aggregate demand by area and time. Keep delivery completion information out of features used before completion. These are example tasks, not requirements.

Possible investigation. How do old operating patterns and a growing share of inaccurate or unrepresentative records change the algorithm's estimates? Separate record accuracy from representation. A minority area or unusual delivery is not automatically low quality.

Access and limits. Use the Files tab and a city CSV from the delivery or pickup folder, not the aggregate viewer, which currently reports a schema mismatch. Check actual headers against the data dictionary. Courier data support a logistics analogy, not validation of autonomous-robot performance. [1]

Rights check for teams. The repository metadata lists Apache-2.0, while the author paper states CC BY-NC 4.0. Keep both references visible and resolve the applicable dataset terms before any redistribution. This package does not assign a blanket license to LaDe. [1, 2]

2. CMU: package-delivery quadcopter data

KiltHub dataset, files, and metadata

What is available. CMU provides 196 parameter-varied flights and 13 ancillary/hover recordings, approximately 243 MB. Measurements include position, wind, current, and voltage. The experiment varies payload, altitude, and speed. The measurements were synchronized at approximately 5 Hz. CC BY 4.0 is listed. [3]

Suggested start. Use complete flights to build an energy-use or performance predictor. Define the decision point and derive any outcome consistently. Separate entire flights across development and evaluation. Thousands of adjacent sensor readings are not thousands of independent flights.

Possible investigation. What happens when historical flight conditions no longer represent the task mix, or when sensor values become stale or systematically unreliable? A documented progression of operating conditions may be simulated, but do not claim that long-term battery aging or fleet drift was measured in these recordings.

Access and limits. Use the dataset's Download all or file controls and inspect the included documentation. It is a research-flight collection, not an Amazon or Walmart commercial drone-fleet history. Its sensor density is useful, but the number of independent flights limits the claims a team can make. [3]

Dataset cards: sea and route learning

3. MarineCadastre: AIS vessel traffic

Bulk-download hub | NOAA access and format guidance

What is available. NOAA's guidance points to vessel-traffic archives from 2009 onward, with formats varying by year. Recent options include CSV and GeoParquet. AIS data contain reported vessel positions and characteristics. Field names and cleaning differ by product. [6, 7]

Suggested start. Select several weeks or months around one port or coastal region. Use broadcast-point records rather than a transit-count map when the experiment needs observation timing. A team can define a documented arrival proxy or predict future position. Neither should be mislabeled as a verified cargo-delivery outcome.

Possible investigation. How do delayed reports, missing updates, or a high-volume unreliable source affect predictions over a voyage or successive operating periods? A held-back future position may support evaluation, but it must not become a live input.

Access and limits. Use bulk files as the primary pathway. A custom-ordering outage notice for AccessAIS was recorded on 8 October, and current service status has not been confirmed, so check service status before relying on custom orders. NOAA also documents extra cleaning in GeoParquet products, so state which product is used and which defects are observed versus experimentally introduced. Historical AIS is not labeled autonomous-vessel control data. Review the selected product's metadata and terms before redistribution. [6–8]

4. Amazon: Last Mile Routing Research Challenge

Dataset registry and access | Data structures

What is available. The collection includes route-, stop-, and package-level information for 9,184 historical driver routes from 2018 across five U.S. metropolitan areas. It separates 6,112 training routes and 3,072 evaluation routes. Data include stop sequences, time windows, planned service times, and historical travel-time estimates. [4, 5]

Suggested start. Select one delivery-station group or a manageable set of complete routes and evaluate a route/sequence recommendation task. Verify dates before describing a split as chronological. Do not treat a human driver's route as an unquestionable optimum.

Possible investigation. How does a learner's recommendation change when older travel assumptions or an increasingly unrepresentative mix of historical routes dominate its evidence? Preserve full routes and justify the evaluation objective. Do not present injected defects as documented Amazon operational failures.

Access and limits. The AWS registry supplies anonymous S3 access instructions. No AWS account is required. Use the listed training/evaluation folders and schema. The source is fixed historical ground-delivery data, not Prime Air telemetry.

Rights check for teams. CC BY-NC 4.0 is listed. Confirm permitted competition and publication uses under that license before redistribution or public write-ups. [4, 5]

What to measure and report

Measurement	What to report
Primary task performance	A task-appropriate metric with units, direction, denominators, and results by time window. Examples: mean absolute prediction error, energy-estimation error, or a justified route objective.
Degradation and recovery	The same metric for reference, stale, low-quality, and combined conditions, without and with the countermeasure. Show when performance changes and what is recovered.
False positives / negatives	For a detector or classifier, define the positive class and report confusion counts and rates. For a pure regressor, use suitable error/tail metrics. Do not invent an arbitrary classification task.
Usefulness and coverage	Report outputs withheld, data discarded, decisions deferred, invalid outputs, and subgroup or rare-event effects when relevant. Rejecting everything is not restored performance.
Time, workload, and trust	Separate compute latency from human review/correction effort. State how understanding or trust was examined and the limits of the test.
Cost and uncertainty	Disclose tools/compute costs, repeat-run variability or other uncertainty checks, residual failures, and practical deployment assumptions.

The key finding. Show the chain from data age, data quality, and source volume to algorithm behavior to real-world consequence. Then show which part of that chain your countermeasure changes. That is the heart of the case, whatever the transport mode.

Sources and verification notes

Bracketed references below support factual dataset descriptions and historical examples.

[1] Cainiao-AI. LaDe dataset card and downloadable files.
Dataset scale, fields, CSV access, and repository metadata. The landing page currently reports a viewer/schema error. Use individual CSV files.

[2] Wu et al. LaDe: The First Comprehensive Last-mile Delivery Dataset from Industry.
Author paper. Its noncommercial license statement differs from the repository's Apache-2.0 metadata.

[3] Carnegie Mellon University. Data Collected with Package Delivery Quadcopter Drone.
196 parameter-varied flights plus 13 ancillary/hover recordings. Synchronized measurements. Approximately 243 MB. CC BY 4.0.

[4] Amazon. 2021 Last Mile Routing Research Challenge Dataset, AWS Open Data Registry.
9,184 routes. 6,112 training and 3,072 evaluation routes. Public S3 access. CC BY-NC 4.0.

[5] MIT CAVE Lab. Amazon routing challenge data structures.
Route, stop, package, time-window, sequence, and historical travel-time field definitions.

[6] NOAA Office for Coastal Management. AIS vessel-traffic guidance.
Bulk archives, formats, GeoParquet documentation, and data-quality caveats.

[7] MarineCadastre. Vessel Traffic download hub.
Primary entry point for bulk vessel-traffic downloads and available products.

[8] MarineCadastre. AccessAIS service.
A custom-ordering outage notice was recorded on 8 October 2026, and current service operation was not verified on the 15 October recheck. Use bulk access and check service status before relying on custom orders.

[9] NIST. Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations, AI 100-2 E2025.
Technical distinction between poisoning, evasion, and other adversarial mechanisms.

[10] NIST. AI Risk Management Framework.
Optional voluntary reference for risk, accountability, and governance discussions. Not an operational authorization.

[11] Jeffrey Dastin, Reuters. Amazon scraps secret AI recruiting tool that showed bias against women, October 2018.
Original reporting on an experimental recruiting system and unrepresentative historical training data.

[12] McAfee Labs. Model Hacking ADAS to Pave Safer Roads for Autonomous Vehicles, February 2020.
Original researchers' account of a modified speed-limit sign being misclassified by the tested Tesla/Mobileye system.

[13] Eykholt et al. Robust Physical-World Attacks on Deep Learning Models.
Research on deliberately modified road signs. Distinct from accidental degradation and not evidence of Tesla stop-sign behavior.

[14] See, Judi. Human Readiness Levels Explained.
Defines the nine human readiness levels in the scale, explains their meaning, and illustrates their application.

Access status is not a guarantee of future availability. This package contains no full third-party dataset, operational prototype, or source-data quality certification. Source terms and product-specific metadata remain controlling.